

9. Joint distributions I.

I. Basic concepts

Let X and Y be random variables defined on the same probability space. ($\Omega, \mathcal{F}, P$)

def. We call $F(a, b) = P(\{X < a\} \cap \{Y < b\})$ the joint cumulative distribution function.

def. We call $F_X(a) = P(X < a) = \lim_{b \rightarrow \infty} F(a, b)$ the marginal distribution function of X .

Similarly, $F_Y(b) = P(Y < b) = \lim_{a \rightarrow \infty} F(a, b)$ is the marginal distr. func. of Y .

Note. $P(X < a) = P(\{X < a\} \cap \bigcup_{b \in \mathbb{N}} \{Y < b\}) = P(\bigcup_{b \in \mathbb{N}} (\{X < a\} \cap \{Y < b\})) \stackrel{\text{continuity}}{=} \lim_{b \rightarrow \infty} P(\{X < a\} \cap \{Y < b\}) = \lim_{b \rightarrow \infty} F(a, b)$

Prop. $P(\{a \leq X < c\} \cap \{b \leq Y < d\}) = F(c, d) - F(a, d) - F(c, b) + F(a, b)$

proof.

a sieve-formula-like argument

$$\begin{aligned}
 P(\{a \leq X < c\} \cap \{b \leq Y < d\}) &= P(\{a \leq X\} \cap \{X < c\} \cap \{b \leq Y\} \cap \{Y < d\}) \\
 &= P(\{a \leq X\} \cap \{X < c\} \cap \{b \leq Y\}) - P(\{a \leq X\} \cap \{X < c\} \cap \{Y < b\}) \\
 &= P(\{a \leq X < c\} \cap \{Y < d\}) - P(\{a \leq X < c\} \cap \{Y < b\}) \\
 &= P(\{X < c\} \cap \{Y < d\}) - P(\{X < a\} \cap \{Y < d\}) - \\
 &\quad - P(\{X < c\} \cap \{Y < b\}) + P(\{X < a\} \cap \{Y < b\}) \\
 &= F(c, d) - F(a, d) - F(c, b) + F(a, b)
 \end{aligned}$$

def. We say that X and Y are jointly discrete random variables if both are discrete over the same probability space.

Writing $\{x_i\}_{i \in I}$, $\{y_j\}_{j \in J}$ for the possible values of X and Y respectively, their joint weight function is $p(x_i, y_j) = P(X = x_i, Y = y_j)$

Marginal weights for X : $p_X(x_i) = \sum_{j \in J} P(X = x_i, Y = y_j) = \sum_{j \in J} p(x_i, y_j)$

Marginal weights for Y : $p_Y(y_j) = \sum_{i \in I} P(X = x_i, Y = y_j) = \sum_{i \in I} p(x_i, y_j)$

Example: There are 3 red, 4 blue and 5 green balls in an urn.

We pick 3 balls randomly, without replacement.

Let X = number of picked red balls, Y = number of picked blue balls

Determine the joint distribution of X and Y !

$Y \backslash X$	0	1	2	3	← number of reds
0	$\frac{\binom{5}{3}}{\binom{12}{3}}$	$\frac{\binom{3}{1}\binom{5}{2}}{\binom{12}{3}}$	$\frac{\binom{3}{2}\binom{5}{1}}{\binom{12}{3}}$	$\frac{1}{\binom{12}{3}}$	$\frac{\binom{8}{3}}{\binom{12}{3}}$
1	$\frac{\binom{4}{1}\binom{5}{2}}{\binom{12}{3}}$	$\frac{\binom{4}{3}\binom{5}{1}}{\binom{12}{3}}$	$\frac{\binom{4}{2}\binom{5}{2}}{\binom{12}{3}}$	0	$\frac{\binom{4}{1}\binom{8}{2}}{\binom{12}{3}}$
2	$\frac{\binom{4}{2}\binom{5}{1}}{\binom{12}{3}}$	$\frac{\binom{3}{2}\binom{4}{1}}{\binom{12}{3}}$	0	0	$\frac{\binom{4}{2}\binom{8}{4}}{\binom{12}{3}}$
3	$\frac{\binom{4}{3}}{\binom{12}{3}}$	0	0	0	$\frac{\binom{4}{3}}{\binom{12}{3}}$
↑ number of blues	$\frac{\binom{3}{3}}{\binom{12}{3}}$	$\frac{\binom{3}{1}\binom{3}{2}}{\binom{12}{3}}$	$\frac{\binom{3}{2}\binom{3}{1}}{\binom{12}{3}}$	$\frac{1}{\binom{12}{3}}$	$\frac{P_X}{P_Y}$

$$P(X=i, Y=j) = \begin{cases} \frac{\binom{3}{i}\binom{4}{j}\binom{5}{3-i-j}}{\binom{12}{3}} & 0 \leq i \\ & 0 \leq j \\ & i+j \leq 3 \\ 0 & \text{else} \end{cases}$$

the marginal distributions,

$$P_X(i) = \frac{\binom{3}{i}\binom{9}{3-i}}{\binom{12}{3}}, \quad 0 \leq i \leq 3$$

$$P_Y(j) = \frac{\binom{4}{j}\binom{8}{3-j}}{\binom{12}{3}}, \quad 0 \leq j \leq 3$$

def: We make n -many independent trials, each has r -many different possible outcomes. Write p_i for the probability of the i th outcome, and let X_i be the number of trials with outcome i . The joint distribution of $\{X_i\}_{i=1}^r$ is called multinomial distribution.

For $n_1 + \dots + n_r = n$ and $n_i > 0$,

$$P(X_1=n_1, \dots, X_r=n_r) = \binom{n}{n_1, \dots, n_r} p_1^{n_1} \dots p_r^{n_r}$$

def: We say that a pair of random variables (X, Y) is jointly absolutely continuous if there exists a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$\forall f(x, y) \geq 0 \quad \forall \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \quad \forall \text{ for any } C \subseteq \mathbb{R}^2, P((X, Y) \in C) = \iint_C f(x, y) dx dy$$

The function f is called the joint density function of (X, Y) .

Note: $F(a, b) = P(\{X < a\} \cap \{Y < b\}) = \int_{-\infty}^a \int_{-\infty}^b f(x, y) dx dy \Rightarrow \frac{\partial^2 F(a, b)}{\partial a \partial b} = \frac{\partial^2 F(a, b)}{\partial b \partial a} = f(a, b)$

def: We call the functions $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ the marginal density of X and $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$ the marginal density of Y .

Example 1 (Uniform distribution)

Let $C \subseteq \mathbb{R}^d$ be a set with finite d -dimensional volume. We say that the tuple of random variables $\underline{X} = (X_1, \dots, X_d)$ is uniformly distributed on C if

$$P(\underline{X} \in A) = \frac{|A|}{|C|}, \quad \forall A \subseteq C \quad f(\underline{x}) = \begin{cases} 1/|C|, & \underline{x} \in C \\ 0, & \text{otherwise} \end{cases}$$

Example 1 The joint density of X and Y is

$$f(x, y) = \begin{cases} e^{-x/y} e^{-y}/y, & x, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

What is the marginal density function of Y ?

It's indeed a density

$$\int_0^\infty \int_0^\infty e^{-x/y} \frac{e^{-y}}{y} dx dy = \int_0^\infty \left[-e^{-x/y} e^{-y} \right]_{x=0}^{x=\infty} dy = \int_0^\infty e^{-y} dy = \left[-e^{-y} \right]_{y=0}^{y=\infty} = 1.$$

$$\underline{f_Y}(y) = \int_{-\infty}^\infty f(x, y) dx = \int_0^\infty e^{-x/y} \frac{e^{-y}}{y} dx = \left[-e^{-x/y} e^{-y} \right]_{x=0}^{x=\infty} = e^{-y} \Rightarrow Y \sim \text{Exp}(1)$$

Example 1 The joint density function of X and Y is

$$f(x, y) = \begin{cases} c, & x^2 + y^2 \leq R^2 \\ 0, & \text{otherwise} \end{cases} \Rightarrow (X, Y) \text{ is uniformly distributed on a circle around } (0, 0) \text{ with radius } R$$

$$c = ? \quad f_X(x) = ? \quad f_Y(y) = ?$$

Let $Z = \sqrt{X^2 + Y^2}$, what are the distribution function and expected value of Z ?

$$1 = \int_{-\infty}^\infty \int_{-\infty}^\infty f(x, y) dx dy = \int_{\text{area of a circle}} R^2 \pi \cdot c \Rightarrow \boxed{c = \frac{1}{R^2 \pi}}$$

$$\underline{f_X}(x) = \int_{-\infty}^\infty f(x, y) dy = \begin{cases} \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} \frac{1}{R^2 \pi} dy, & x \in [-R, R] \\ 0, & \text{otherwise} \end{cases} = \begin{cases} \frac{2\sqrt{R^2-x^2}}{R^2 \pi}, & x \in [-R, R] \\ 0, & \text{otherwise} \end{cases}$$

By symmetry $\underline{f_Y}(y) = \underline{f_X}(y)$.

$$\underline{P(Z < a)} = \underline{P(X^2 + Y^2 < a^2)} = \begin{cases} 1, & a > R \\ \frac{a^2 \pi}{R^2 \pi}, & 0 \leq a \leq R \\ 0, & a \leq 0 \end{cases}$$

$Z = \sqrt{X^2 + Y^2}$
 (X, Y) is inside a circle around $(0, 0)$ of radius a

$$\Rightarrow \underline{f_Z}(a) = \frac{2a}{R^2}$$

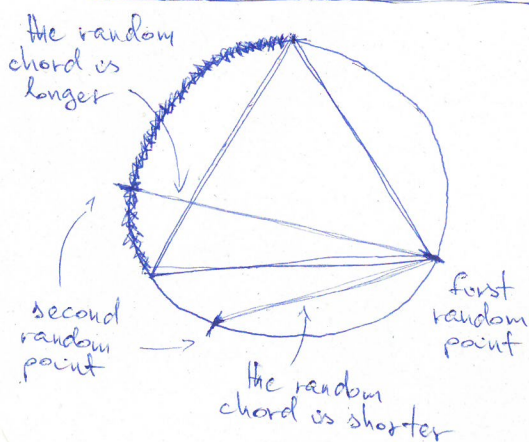
$$\underline{E[Z]} = \int_0^R z \frac{2z}{R^2} dz = \left[\frac{2}{3} \frac{z^3}{R^2} \right]_{z=0}^{z=R} = \frac{2}{3} R$$

II. Bertrand's paradox

What is the probability that a random chord in the unit circle is longer than the side length of a regular triangle inscribed the circle?

Bertrand showed that depending on our method of choosing a random chord there are more than just one correct (and different) solutions for this problem.

a) "Random endpoints" method:



Choose two random points on the circumference of the circle, and draw the chord joining them.

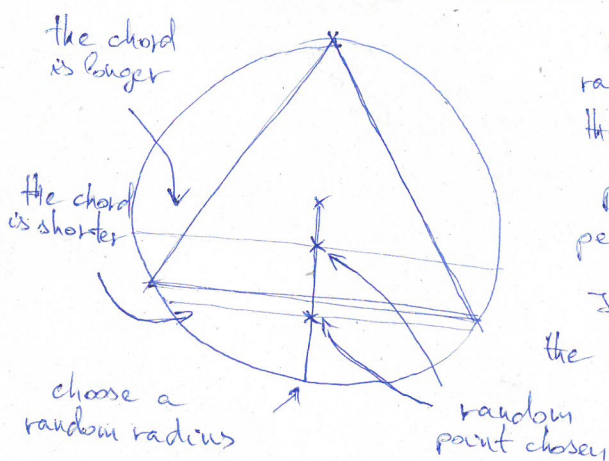
Draw the triangle in a way, that its vertex is the first random point.

If the second point lies on the arc between the endpoints of the side of the triangle opposing the first point, then the chord is longer than the side length of the triangle.

The length of this arc is $\frac{1}{3}$ of the circumference of the circle.

$$\Rightarrow \boxed{p = \frac{1}{3}}$$

b) "Random radial point" method:



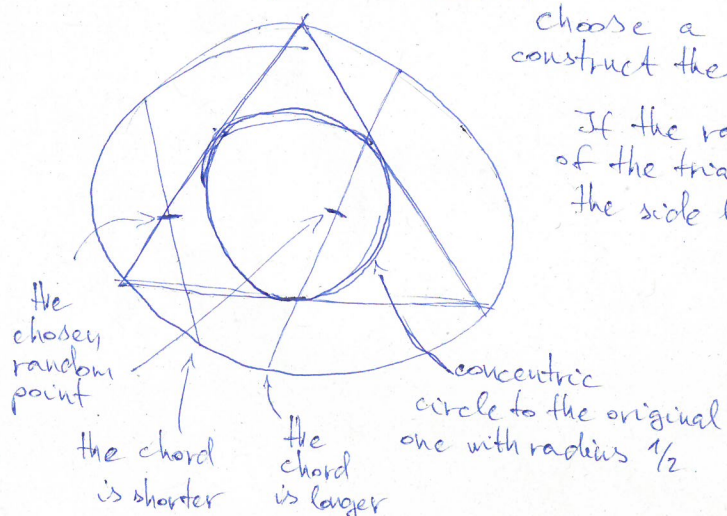
Choose a radius of the circle, then choose a random point on it. Now, construct the chord through this point and perpendicular to the radius.

Draw the triangle in a way, that the radius is perpendicular to the side it intersects.

If the chosen point is inside the ~~(chord)~~ triangle, then the chord is longer than the side length.

$$\Rightarrow \boxed{p = \frac{1}{2}}$$

c) "Random midpoint" method:



choose a random point inside the circle, and construct the chord with the chosen point as its midpoint.

If the random point is inside of the inscribed circle of the triangle, then the chord is longer than the side length.

$$\boxed{p = \frac{\text{area of (big) circle}}{\text{area of (small) circle}} = \frac{1}{4}}$$

III. Transformations

Let $\underline{X} = (X_1, \dots, X_d)$ be a tuple of jointly absolutely continuous random variables with joint density f . That is

$$f: \mathbb{R}^d \rightarrow \mathbb{R}, \quad \forall \underline{x} \in \mathbb{R}^d, f(\underline{x}) \geq 0, \quad \int_{\mathbb{R}^d} f(\underline{x}) d\underline{x} = 1, \quad P(\underline{X} \in A) = \int_A f(\underline{x}) d\underline{x}$$

Let $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ be such that

- 1) g is one-to-one;
- 2) g has continuous partial derivatives;

$$3) |\underline{J}| = \begin{vmatrix} \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_d}{\partial x_1} \\ \vdots & & \vdots \\ \frac{\partial g_1}{\partial x_d} & \dots & \frac{\partial g_d}{\partial x_d} \end{vmatrix} \neq 0 \text{ on } \mathbb{R}^d.$$

Then g is invertible and $\underline{Y} = g(\underline{X}) \Rightarrow f_{\underline{Y}}(\underline{y}) = \frac{f_{\underline{X}}(g^{-1}(\underline{y}))}{|\det(\underline{J})|}$

IV. Independent random variables

We say that the random variables $X_1, \dots, X_n$ are independent if for every $A_1, \dots, A_n \subseteq \mathbb{R}$

$$P(\{X_1 \in A_1\} \cap \{X_2 \in A_2\} \cap \dots \cap \{X_n \in A_n\}) = P(X_1 \in A_1) P(X_2 \in A_2) \dots P(X_n \in A_n)$$

We say that a countable family of random variables are independent if every finite subcollection is independent.

Remark: For random variables $X_1, X_2, \dots$ it is enough to check that X_2 is independent of X_1 ; X_3 is independent of X_2 and X_1 ; and so on X_n is independent of $X_{n-1}, \dots, X_2, X_1$ for each $n \in \mathbb{N}$.

Prop: The following statements are equivalent,

a) X and Y are independent

$$b) F(x, y) = F_X(x) F_Y(y)$$

$$c) \begin{cases} f(x, y) = f_X(x) f_Y(y) & \text{in the absolutely continuous case} \\ p(x, y) = p_X(x) p_Y(y) & \text{in the discrete case} \end{cases}$$

Example: Show that if $X \sim \text{Uni}[a, b]$ and $Y \sim \text{Uni}[c, d]$, then $(X, Y) \sim \text{Uni}([a, b] \times [c, d])$ and X and Y are independent.

$$f(x, y) = \begin{cases} \frac{1}{(b-a)(d-c)} & , a \leq x \leq b \text{ \& } c \leq y \leq d \\ 0 & , \text{otherwise} \end{cases}$$

$$f_X(x) = \begin{cases} \frac{1}{(b-a)} & , a \leq x \leq b \\ 0 & , \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{(d-c)} & , c \leq y \leq d \\ 0 & , \text{otherwise} \end{cases}$$

Note: If $(X, Y) \sim \text{Uni}(C)$ and X, Y are independent, then C is an axis parallel rectangle.

Example 1: On a given day, the number of customers who enters the post office follows a Poisson distribution with parameter λ . A customer is male with probability p and female with probability $1-p$. What are the marginal weight functions of the number of male and female customers?
Show that they are independent!

X = number of male customers, Y = number of female customers

$$P(X+Y=n) = e^{-\lambda} \frac{\lambda^n}{n!}$$

$$\begin{aligned} P(X=i, Y=j) &= P(X=i, Y=j | X+Y=i+j) P(X+Y=i+j) = \\ &= \binom{i+j}{i} p^i (1-p)^j e^{-\lambda} \frac{\lambda^{i+j}}{(i+j)!} = e^{-\lambda p} \frac{(p\lambda)^i}{i!} e^{-\lambda(1-p)} \frac{((1-p)\lambda)^j}{j!} \end{aligned}$$

$$\Rightarrow P(X=i) = \sum_{j=0}^{\infty} P(X=i, Y=j) = \sum_{j=0}^{\infty} e^{-\lambda p} \frac{(p\lambda)^i}{i!} e^{-\lambda(1-p)} \frac{((1-p)\lambda)^j}{j!} = e^{-\lambda p} \frac{(p\lambda)^i}{i!}$$

weights of $Poi((1-p)\lambda)$

Similarly

$$P(Y=j) = e^{-(1-p)\lambda} \frac{((1-p)\lambda)^j}{j!}$$

$$P(X=i, Y=j) = P(X=i)P(Y=j)$$

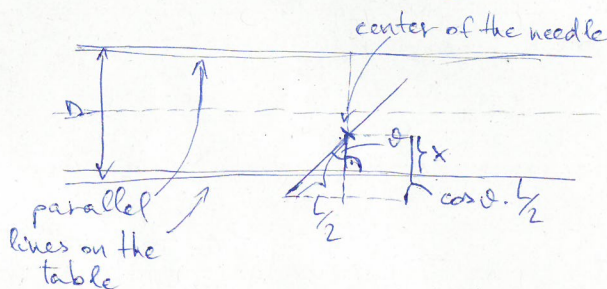
$\Rightarrow X \sim Poi(\lambda p), Y \sim Poi((1-p)\lambda)$
and they are independent

Example 1: (Buffon's needle problem)

A table is ruled by parallel lines of distance D . A needle with length $L \leq D$ is randomly thrown on the table. What is the probability that the needle intersects a line?

X = distance of the center of the needle from the closest parallel line
 $X \sim Uni[0, D/2]$

θ = angle of the needle



$$\begin{aligned} P(\text{the needle intersects a line}) &= \\ &= P(\cos \theta \cdot \frac{L}{2} \geq X) = \int_0^{\pi/2} \int_0^{\frac{L}{2} \cos \theta} \frac{4}{\pi D} dx d\theta = \\ &= \int_0^{\pi/2} \frac{2L \cos \theta}{\pi D} d\theta = \left[\frac{2L}{D\pi} \sin \theta \right]_{\theta=0}^{\theta=\pi/2} = \frac{2L}{D\pi} \end{aligned}$$

def. Let $\underline{A}$ be a $d \times d$ symmetric positive definite matrix ($\forall \underline{x} \neq \underline{0}, \underline{x}^T \underline{A} \underline{x} > 0$) and let $\underline{m} \in \mathbb{R}^d$. We say that the random variable $\underline{X}$ has d -dimensional joint normal distribution with parameters $\underline{m}$ and $\underline{A}$ if its density function is

$$f(\underline{x}) = \frac{\sqrt{\det(\underline{A})}}{\sqrt{(2\pi)^d}} e^{-\frac{1}{2}(\underline{x}-\underline{m})^T \underline{A} (\underline{x}-\underline{m})}$$

Prop. A random variable $\underline{X}$ with d -dimensional joint normal distribution is a linear transformation of d independent standard normal variables.

proof:

A is symmetric and positive definite $\Rightarrow$

$\exists \underline{P}$ orthogonal matrix, $\exists \underline{D}$ diagonal matrix with strictly positive diagonal elements with determinant = 1

$$\underline{A} = \underline{P} \underline{D} \underline{P}^T, \text{ and let } g(\underline{x}) := \sqrt{\underline{D}} \underline{P}^T (\underline{x} - \underline{m})$$

Since $\underline{A} = \underline{P} \underline{D} \underline{P}^T = (\underline{P}^T)^T \sqrt{\underline{D}} \sqrt{\underline{D}} \underline{P}^T = (\sqrt{\underline{D}} \underline{P}^T)^T \sqrt{\underline{D}} \underline{P}^T$ we have

$$f_{\underline{X}}(\underline{x}) = \frac{\sqrt{\det(\underline{D})}}{\sqrt{2\pi}^d} e^{-\frac{1}{2} g(\underline{x})^T g(\underline{x})}$$

Let $\underline{Y} = g(\underline{X})$, then $f_{\underline{Y}}(\underline{y}) = \frac{f_{\underline{X}}(g^{-1}(\underline{y}))}{|\det(\underline{J})|} = \frac{\sqrt{\det(\underline{D})}}{\sqrt{2\pi}^d} e^{-\frac{1}{2} \underline{y}^T \underline{y}} \cdot \frac{1}{\det(\sqrt{\underline{D}})} \checkmark$

density of transformed variable

$$= \frac{1}{\sqrt{2\pi}^d} e^{-\frac{1}{2} \underline{y}^T \underline{y}} = \prod_{i=1}^d \frac{1}{\sqrt{2\pi}} e^{-\frac{y_i^2}{2}}$$

$\underline{y} = (y_1, \dots, y_d)$ density of $N(0,1)$

$\Rightarrow$ The random variables $(Y_1, \dots, Y_d)$ are independent standard normals.

$\Rightarrow \underline{X} = g^{-1}(\underline{Y}) = \underline{P} (\sqrt{\underline{D}})^T \underline{Y} + \underline{m}$ is a linear transformation of independent standard normal random variables.